

CO3_AT-1 Case Studies (1–5)

Case Study 1 – Employee Salaries

Given Data:

4, 5, 5, 6, 7, 8, 8, 9, 30

Q1. Find the mean, median, and mode.

Formula:

$$\text{Mean} = \Sigma x/n$$

$$\Sigma x = 4+5+5+6+7+8+8+9+30 = 82$$

$$n = 9$$

$$\text{Mean} = 82/9 = 9.11 \text{ lakhs}$$

Median:

Ordered data = 4,5,5,6,7,8,8,9,30

$$\text{Median position} = (9+1)/2 = 5\text{th value} = 7$$

Mode:

5 appears 2 times

8 appears 2 times

Mode = 5 and 8

Q2. Find the range.

Formula:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$= 30 - 4$$

$$= 26 \text{ lakhs}$$

Q3. Calculate the variance and standard deviation.

Formula:

$$\text{Variance} = \Sigma (x-\mu)^2/n$$

$$\text{SD} = \sqrt{\text{Variance}}$$

$$\text{Mean } \mu = 9.11$$

x	$x-\mu$	$(x-\mu)^2$
4	-5.11	26.12
5	-4.11	16.90
5	-4.11	16.90
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.35

$$\Sigma(x-\mu)^2 = 512.88$$

$$\text{Variance} = 512.88/9 = 56.99$$

$$\text{SD} = \sqrt{56.99} = 7.55$$

Q4. Identify the outlier.

30 lakhs is an outlier because it is much larger than the remaining salaries.

Q5. Which measure of central tendency best represents the typical salary? Explain why

The median is the best measure because the salary ₹30 lakhs is an outlier. This unusually high salary increases the mean (₹9.11 lakhs), making it higher than the salaries of most employees. The median (₹7 lakhs) is not affected by the outlier and better represents the typical employee salary.

Case Study 2 – Website Response Time

Given Data:

120,125,128,130,132,135,140,145,150,300

Q1. Calculate the mean and median. Σx

$$=120+125+128+130+132+135+140+145+150+300=1505$$

$$\text{Mean}=1505/10=150.5 \text{ ms}$$

$$\text{Median}=(132+135)/2=133.5 \text{ ms}$$

Q2. Find the range.

$$\text{Range}=300-120=180 \text{ ms}$$

Q3. Calculate the variance and standard deviation.

Formula:

$$\text{Variance}=\Sigma(x-\mu)^2/n$$

$$\text{Mean}=150.5$$

x	$x-\mu$	$(x-\mu)^2$
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120	-30.5	930.25
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125	-25.5	650.25
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128	-22.5	506.25
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130	-20.5	420.25
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132	-18.5	342.25
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135	-15.5	240.25
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140	-10.5	110.25
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145	-5.5	30.25
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150	-0.5	0.25
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300	149.5	22350.25
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$$\Sigma(x-\mu)^2=25580.50$$

$$\text{Variance}=25580.50/10=2558.05$$

$$\text{SD}=\sqrt{2558.05}=50.58 \text{ ms}$$

Q4. Identify the outlier.

The data is positively (right) skewed due to the value 300.

Q5. Which measure of central tendency best represents the typical salary? Explain why

Detailed Explanation:

The response times range from 120 ms to 300 ms.

Most of the response times are between 120 ms and 150 ms, but one request took 300 ms, which is much larger than the others.

Because the mean includes every value in the calculation, this unusually high response time increases the average response time to 150.5 ms.

The median is 133.5 ms, which better represents the response time experienced by most users.

In web performance analysis, a few unusually slow requests should not represent the overall performance of the website.

Therefore, engineers often prefer the median because it is less affected by extreme values and better reflects the normal response time.

Case Study 3 – Student Performance

Given Data:

42,45,48,50,52,52,55,58,60,62,65,95

Q1.Find Q1, Q2 and Q3.

Lower half: 42,45,48,50,52,52

$$Q1=(48+50)/2=49$$

$$\text{Median}(Q2)=(52+55)/2=53.5$$

Upper half:55,58,60,62,65,95

$$Q3=(60+62)/2=61$$

Q2.Calculate the IQR.

$$IQR=Q3-Q1$$

$$=61-49$$

$$=12$$

Q3. Use the $1.5 \times IQR$ rule to identify any outliers.

$$\text{Lower Limit}=49-1.5 \times 12$$

$$=49-18$$

$$=31$$

$$\text{Upper Limit}=61+18$$

$$=79$$

$$95 > 79$$

Therefore 95 is an outlier.

Q4. Find the mean and median.

$$\Sigma x=684$$

$$\text{Mean}=684/12=57$$

$$\text{Median}=53.5$$

Q5. Explain how the outlier affects the mean.

Detailed Explanation:

The marks of most students range between 42 and 65, while one student scored 95.

The score 95 is an outlier, since it is much higher than the rest of the marks.

When calculating the mean, every mark contributes equally.

Because of the very high score of 95, the total marks increase significantly, causing the mean (57) to become higher than the performance of most students.

The median (53.5) is not greatly affected because it depends only on the middle values.

If the outlier were removed, the mean would decrease and become closer to the median.

Case Study 4 – E-Commerce Orders

Given Data:

18,20,22,22,24,25,26,28,30,35

Q1. Calculate the mean, median, and mode.

$$\Sigma x = 250$$

$$\text{Mean} = 250/10 = 25$$

$$\text{Median} = (24 + 25)/2 = 24.5$$

$$\text{Mode} = 22$$

Q2. Find the variance and standard deviation.

$$\text{Mean} = 25$$

x	$x - \mu$	$(x - \mu)^2$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

$$\Sigma(x - \mu)^2 = 228$$

$$\text{Variance} = 228/10 = 22.8$$

$$\text{SD} = \sqrt{22.8} = 4.77$$

Q3. Calculate the coefficient of variation if the mean and standard deviation are known.

$$CV=(SD/Mean)\times 100$$

$$=(4.77/25)\times 100$$

$$=19.08\%\approx 19.1\%$$

Q4. What does the standard deviation tell the company?

Orders deviate by about 4.77 from the average.

Q5 Would you describe the order data as highly or moderately variable?

Detailed Explanation:

The average number of daily orders is 25, and the standard deviation is approximately 4.77.

The Coefficient of Variation (CV) is calculated as

$$CV=\frac{SD}{Mean}\times 100$$

$$CV=\frac{4.77}{25}\times 100=19.1\%$$

A CV below about 20% generally indicates moderate variability, meaning that the daily orders do not fluctuate drastically.

Most of the daily orders are close to the average, and there are no extremely unusual values.

This indicates that customer demand is relatively stable from day to day.

Case Study 5 – Comparing Two Data Science Models

Given Data:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

Q1. Calculate the mean error for both models.

Model A:

$$\Sigma x = 32$$

$$\text{Mean} = 32/8 = 4$$

Model B:

$$\Sigma x = 37$$

$$\text{Mean} = 37/8 = 4.625$$

Q2. Calculate the mean error for both models.

Model A ($\mu = 4$)

$$(x - \mu)^2:$$

$$4, 1, 1, 0, 0, 1, 1, 4$$

$$\Sigma = 12$$

$$\text{Variance} = 12/8 = 1.5$$

Model B ($\mu = 4.625$)

$$\Sigma(x - \mu)^2 = 73.875$$

$$\text{Variance} = 73.875/8 = 9.23$$

Q3. Calculate the mean error for both models.

Model A:

$$\text{SD} = \sqrt{1.5} = 1.22$$

Model B:

$$SD = \sqrt{9.23} = 3.04$$

Q4. Calculate the mean error for both models.

Model A is more consistent because it has the lower standard deviation.

Q5. Calculate the mean error for both models.

Detailed Explanation:

No, a lower mean error alone is not enough to conclude that a model is better.

When evaluating machine learning models, two important factors are considered:

1. Mean Error

- * Measures the average prediction error.
- * A smaller value generally indicates better accuracy.

2. Standard Deviation (or Variance)

- * Measures how consistent the prediction errors are.
- * A smaller standard deviation means the model produces more reliable predictions.

A model may have a low average error but produce very inconsistent predictions, resulting in a large standard deviation.

Another model may have a slightly higher average error but much smaller variation, making it more dependable in real-world applications.

In this case:

* Model A

- * Mean Error = 4
- * Standard Deviation = 1.22

* Model B

- * Mean Error = 4.625
- * Standard Deviation = 3.04

Model A has both a lower mean error and a much lower standard deviation, making it the better model.